

OPTIMAL ECOG COVERAGE FOR EPILEPSY SURGERY

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ABSTRACT. Resective surgery can free a patient with drug-resistant focal epilepsy from seizures. When an intracranial recording guides it, the resection can only be planned around what the recording’s contacts have seen. We treat the placement of the recording contacts as an optimisation problem on the patient’s cortical surface: every point of the territory suspected of generating the seizures should lie within a footprint radius r of at least k contacts, measured along the folded surface. This holds exactly when the k -th covering radius ρ_k of the territory is below r . An onset zone of any size and position inside the territory is then seen whole, and it stays seen whole after any $k - 1$ contacts fail. On a 3777 mm^2 parietal territory of the colin27 template brain, the documented 8×8 subdural grid leaves 55.1% of the territory unseen at $r = 8 \text{ mm}$, and an onset zone of radius up to 13.9 mm can lie in it and be missed entirely. The same 64 contacts placed by covering radius leave 1.0 mm^2 unseen, and 66 contacts leave nothing. A contact array confined to the gyral crowns, as a subdural sheet is, cannot reach this at any contact count: some point of the territory is 12.219 mm from every crown. After 8 random contact failures, double coverage (112 contacts at 9 mm) still sees a 5 mm onset zone whole in 99.9% of its possible positions on average, where single coverage falls to 74.2%. Complete coverage is what lets a resection planned on the recording contain the onset zone by design; it is not sufficient for seizure freedom, and this paper reports no patient outcome. We state the protocol that would test it.

1. INTRODUCTION

Epilepsy is drug-resistant when two appropriately chosen and tolerated antiseizure medications have failed [21], and a substantial minority of patients reach that point [22]. For focal drug-resistant epilepsy, resective surgery is the treatment that can end the seizures: in randomised trials it outperformed continued medical therapy [33, 9], and seizure freedom, Engel class I or ILAE class 1 [10, 34], is its stated goal [20]. Surgery aims at the epileptogenic zone, the cortex whose removal is needed for seizure freedom, and when intracranial recording is needed, a central estimate of it is the seizure onset zone that recording shows [27].

When the non-invasive evaluation (semiology, scalp video-EEG, MRI, PET, SPECT, MEG) does not settle where seizures begin, contacts are implanted: subdural grids and strips for electrocorticography (ECoG), or depth leads for stereo-EEG [19, 26]. The implantation is hypothesis driven. It samples a territory X that the evaluation says should contain the onset zone, and what it does not sample it cannot localise (Fig. 1). An onset zone that falls between contacts is missed or truncated, and the resection is then planned around a false or partial picture, or not at all. Larger implantations carry more risk [14, 24], so the number of contacts cannot simply be raised until nothing is missed.

This paper asks a narrow question and answers it exactly: given the patient’s cortical surface, the territory X , and a radius r within which a contact is credited with recording an onset, where should n contacts go so that every point of X is seen at least k times, and how small can n be? The answer turns on one number, the k -th covering radius of X measured along the folded surface (Section 2). It gives three things a clinician can use before implantation. First, a guarantee: when the covering

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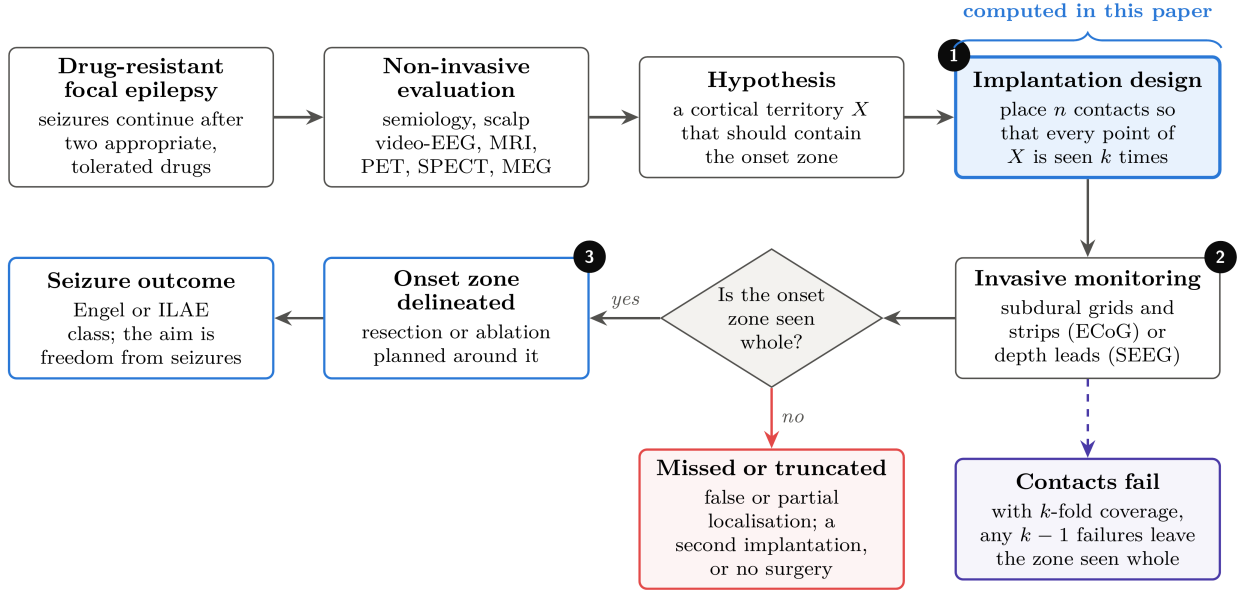


FIGURE 1. Where contact placement enters the surgical pathway. The implantation is designed from a hypothesis, a cortical territory X expected to contain the seizure onset zone (step 1). What the invasive recording can localise is limited to what its contacts see (step 2), and a resection aimed at seizure freedom is planned around the onset zone the recording delineates (step 3). An onset zone that is missed or seen only in part leads to false or partial localisation. With k -fold coverage, any $k - 1$ contact failures during monitoring leave every onset zone in X seen whole (Proposition 3). This paper computes step 1.

radius is below r , an onset zone of any size and position in X is seen whole, and stays seen whole after any $k - 1$ contacts fail. Second, a design rule that meets that guarantee with close to the fewest contacts. Third, an obstruction: contacts confined to the gyral crowns, as on a subdural sheet, cannot meet the guarantee below a radius that is fixed by the folding and not by the number of contacts.

We compute all of this on the colin27 template brain [16, 25], where a documented clinical grid layout [32] can be compared with designed ones, and we extend the coverage question to the one a surgeon asks: for an onset zone of radius s placed anywhere in X , how often is it seen whole, how often is it missed, and how often does it survive contact failures (Section 5)?

What this paper does not claim. Coverage is geometry. Whether a patient becomes seizure-free depends on much more: whether the onset zone recorded is the epileptogenic zone, whether it can be resected without unacceptable deficit, and whether the epilepsy is focal or network-driven [31, 1]. Complete coverage of the territory is what lets a resection planned on the recording contain the onset zone by design, not a promise of cure, and incomplete resection of the recorded onset zone has itself been reported not to predict outcome in one cohort [12]. No patient data and no outcome are used here. Section 6 states the protocol that would test whether optimal coverage changes outcome.

Relation to earlier work. Trajectory planning for stereo-EEG optimises safety and sampling of target structures [6, 30], electrode spacing has been studied for how much signal an array resolves [29], and Jaber et al. test after the fact whether an implantation sampled the epileptogenic zone [18]. Coverage by sensors with a sensing radius is classical in sensor networks [17], its topology can be read from the nerve of the footprints [3, 7], and the k -fold region from the multicover nerve [28]; the

certificate used here as a check follows [2]. The placement rule is farthest-point sampling, the greedy algorithm for the k -centre problem [13, 15]. What is new is the statement of ECoG placement as a covering-radius problem on the folded surface, the site floor that it implies for subdural sheets, and the onset-zone capture and failure analysis of designed arrays.

2. COVERAGE OF A TERRITORY, AND OF AN ONSET ZONE

Let M be the patient's pial surface, a closed triangulated surface, with the geodesic distance d (shortest paths along the surface). Let $X \subseteq M$ be the target territory, $C \subseteq M$ the sites a contact may occupy, and $P = \{p_1, \dots, p_n\} \subseteq C$ the contacts. A contact sees the points within distance r of it along the surface; its footprint is $B_i = \{x \in M : d(x, p_i) < r\}$. The multiplicity of a point is $m(x) = \#\{i : x \in B_i\}$, and the region seen at least k times is $R_k = \{x : m(x) \geq k\}$, so that $R_1 \supseteq R_2 \supseteq \dots$ (Fig. 2).

Definition 1. For $x \in M$ let $d_k(x, P)$ be the distance from x to its k -th nearest contact. The k -th covering radius of X by P is

$$\rho_k(P) = \max_{x \in X} d_k(x, P).$$

Proposition 2 (coverage is a covering radius). $X \subseteq R_k$ if and only if $\rho_k(P) < r$.

Proof. A point x lies in R_k exactly when at least k contacts are within distance less than r of it, that is, when $d_k(x, P) < r$. This holds for every $x \in X$ exactly when it holds for the maximiser, which exists because X is compact (on a mesh, finite). $\square$

So the whole design problem is to push $\rho_k(P)$ below r with as few contacts as possible, and ρ_k is computed exactly by Dijkstra's algorithm [8], one run per contact.

Onset zones and contact failure. An onset zone is a region $Z \subseteq X$ of unknown position. We say Z is *seen whole* if $Z \subseteq R_1$, and *missed* if $Z \cap R_1 = \emptyset$. Contacts fail during monitoring, through lost contact with the cortex, noise or artefact, and the question is whether Z is still seen whole by the contacts that remain.

Proposition 3 (failure-proof sampling). *Let $Z \subseteq X$ and $q \geq 0$. Every set of at most q failed contacts leaves Z seen whole by the survivors if and only if $Z \subseteq R_{q+1}$. In particular, if $\rho_{q+1}(P) < r$, every onset zone in X , of any size and position, is seen whole after any q contacts fail.*

Proof. If $Z \subseteq R_{q+1}$, a point $x \in Z$ lies in at least $q+1$ footprints, at most q of which have failed, so a surviving footprint contains x . Conversely, if some $x \in Z$ has $m(x) \leq q$, let the $m(x)$ contacts whose footprints contain x fail; that is at most q failures, and x is no longer seen. The last sentence follows from Proposition 2 with $k = q+1$. $\square$

The guarantee is worst case: it holds for every failure set, however unlucky, not merely on average. Section 5 also measures the average under random failure.

Capture of an onset zone of given size. When $X \not\subseteq R_k$ the answer depends on the size and position of the onset zone. Model it as the closed geodesic disk $Z_s(x) = \{y : d(x, y) \leq s\}$ of radius s about a centre x , admissible when $Z_s(x) \subseteq X$, and let A_s be the set of admissible centres. With μ the area measure, define

$$W_k(s) = \frac{\mu\{x \in A_s : Z_s(x) \subseteq R_k\}}{\mu(A_s)}, \quad L(s) = \frac{\mu\{x \in A_s : Z_s(x) \cap R_1 = \emptyset\}}{\mu(A_s)}.$$

If the centre of the onset zone is equally likely to be anywhere it fits in X , $W_1(s)$ is the probability that the array sees it whole, $W_2(s)$ the probability that it still does after any one failure, and $L(s)$ the probability that it is missed entirely. The uniform prior is an assumption of the model. With a patient-specific prior, for instance from the concordance of non-invasive tests, μ is replaced by that prior and nothing else changes.

Proposition 4 (capture by erosion). *$Z_s(x) \subseteq R_k$ if and only if $d(x, M \setminus R_k) > s$, and $Z_s(x) \cap R_1 = \emptyset$ if and only if $d(x, R_1) > s$. Consequently:*

- (1) *if $X \subseteq R_k$ then $W_k(s) = 1$ and $L(s) = 0$ for every $s \geq 0$; conversely $W_k(0) = 1$ gives $\mu(X \setminus R_k) = 0$, and on a mesh, where every vertex carries positive area, $X \subseteq R_k$;*
- (2) *onset zones of every radius below $h(P) = \max_{x \in X} \min\{d(x, R_1), d(x, M \setminus X)\}$, and of none above it, can be missed entirely, and*

$$h(P) \geq \max_{x \in X} \min\{d_1(x, P) - r, d(x, M \setminus X)\}.$$

Proof. A closed disk of radius s about x avoids a set S exactly when every point of S is farther than s from x ; take $S = M \setminus R_k$ and $S = R_1$. For (1), if $X \subseteq R_k$ every admissible disk lies in R_k ; conversely $W_k(0) = \mu(X \cap R_k)/\mu(X)$. For (2), $L(s) > 0$ exactly when some x has $d(x, R_1) > s$ and $d(x, M \setminus X) > s$, that is, when $s < h(P)$. For the bound, if $y \in R_1$ then $d(y, p) < r$ for some contact p , so $d(x, y) \geq d(x, p) - r \geq d_1(x, P) - r$. $\square$

Each of W_k and L is therefore one multi-source Dijkstra run, from the unseen set or from the seen set, followed by a threshold at each s . Part (2) turns the covering radius into a clinical statement. An array whose covering radius exceeds the footprint by δ at a point well inside X can miss an onset zone of radius about δ there entirely, whatever else it records.

The shape of the covered region. Where coverage is not complete, its shape matters. A hole in R_1 , a pocket of unseen cortex surrounded by seen cortex, is exactly where an onset zone can hide while every contact around it records (Fig. 2a). The topology of R_k can be read from which footprints overlap: when every nonempty intersection of footprints is contractible, R_k is homotopy equivalent to the k -shadow of the nerve of the footprints [28, 2], whose Betti numbers (b_0, b_1) count pieces and holes. We use this certificate as an independent check on every design, on the understanding established in earlier work [4]: geodesic footprints on a folded cortex are not always convex, so the certificate is checked against a direct computation on the mesh, and a certificate of $(1, 0)$ says that R_k is one piece without holes, not that it contains X . Containment is decided by Proposition 2.

3. WHY A SUBDURAL SHEET CANNOT REACH THE FOLDS

A subdural grid is a flat sheet laid on the surface of the brain. Its contacts sit on the gyral crowns; the banks and fundi of the sulci can lie a centimetre or more of cortex away along the surface, however close they are in a straight line (Fig. 3). Three elementary facts make this exact and say what can be done instead.

Proposition 5 (site floor). *Put $\rho_C = \max_{x \in X} \min_{c \in C} d(x, c)$. Then $\rho_1(P) \geq \rho_C$ for every finite $P \subseteq C$. So if $r \leq \rho_C$, no array with its contacts in C covers X , at any number of contacts.*

Proof. Let $x \in X$ attain the maximum defining ρ_C . Every $p \in P$ lies in C , so $\min_{p \in P} d(x, p) \geq \min_{c \in C} d(x, c) = \rho_C$, and $\rho_1(P)$ is at least this. The second sentence is Proposition 2. $\square$

Proposition 6 (packing bound). *If $S \subseteq X$ has $d(s, s') \geq 2r$ for all distinct $s, s' \in S$, then every array with $X \subseteq R_1$ has at least $|S|$ contacts.*

Proof. If s and s' lay in one footprint $B(p, r)$, then $d(s, s') \leq d(s, p) + d(p, s') < 2r$. So each footprint contains at most one point of S , and the footprints cover S . $\square$

Proposition 7 (farthest-point placement is within a factor two). *On a mesh, with ρ_1 taken over the vertices of X , let $p_1, p_2, \dots$ be chosen from the vertices of X , each p_{j+1} a vertex of X farthest from $P_j = \{p_1, \dots, p_j\}$. Then for every n and every set Q of n points of M , $\rho_1(P_n) \leq 2\rho_1(Q)$.*

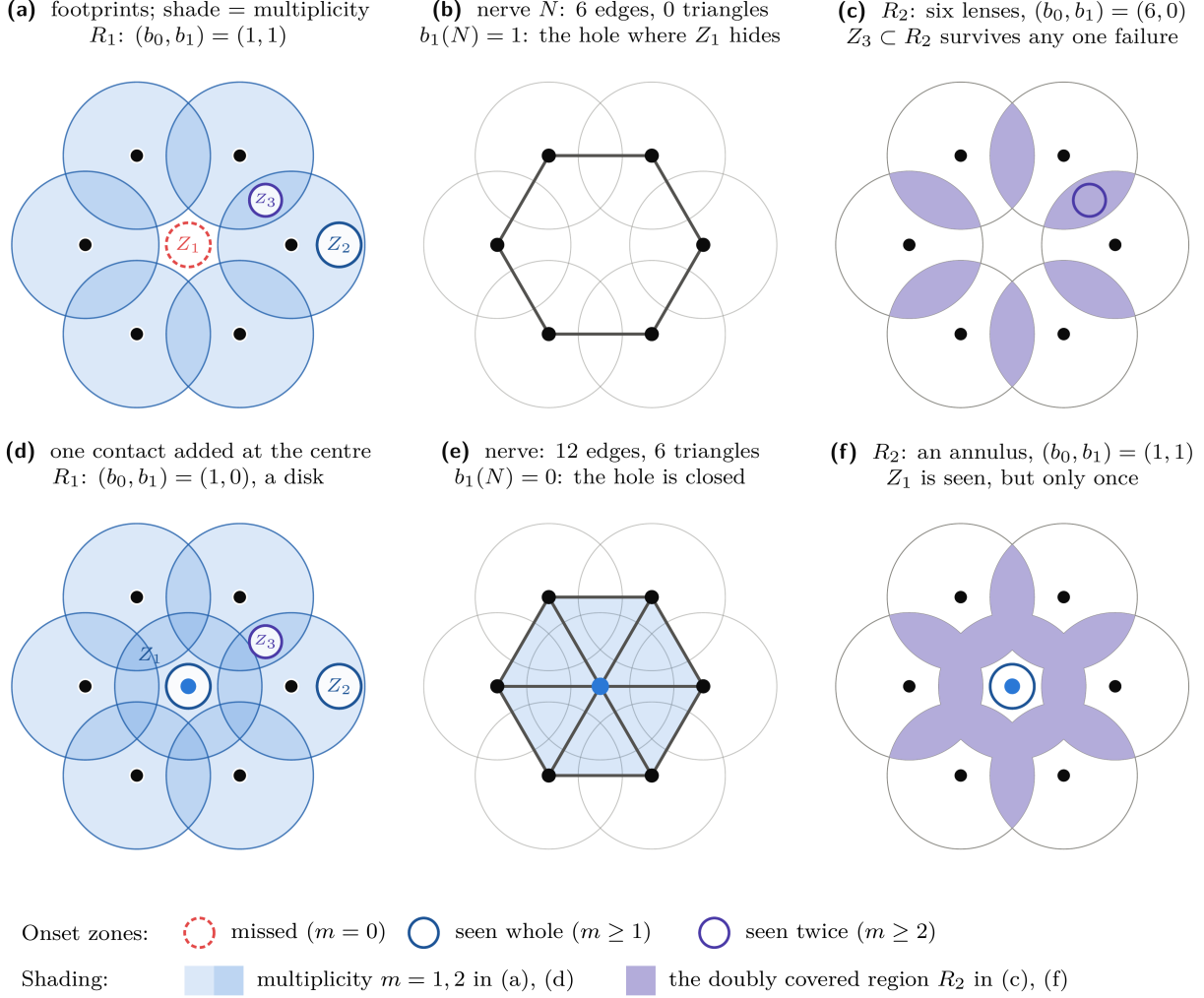


FIGURE 2. Footprints, the nerve, and what they say about an onset zone. Six contacts on a ring, footprints of radius 1 (top), and the same with one contact added at the centre (bottom). (a) The multiplicity m of the footprints. R_1 has a hole, and an onset zone Z_1 inside it is missed although six contacts surround it; Z_2 is seen whole once; Z_3 lies where two footprints overlap. (b) The nerve: an edge for each pair of overlapping footprints, a triangle for each triple with a common point. Its cycle is the hole of (a), found from overlaps alone. (c) The doubly covered region R_2 , six lenses; $Z_3 \subset R_2$ is still seen whole if any one contact fails (Proposition 3). (d)–(f) One contact added at the centre closes the hole: Z_1 is now seen, but only once, since R_2 is an annulus around it. Every statement in the panel titles was checked numerically.

Proof. Since p_{j+1} is a farthest vertex of X from P_j , $d(p_{j+1}, P_j) = \rho_1(P_j)$, and $\rho_1(P_j)$ does not increase with j . So any two of $p_1, \dots, p_{n+1}$ are at least $\rho_1(P_n)$ apart. Send each of these $n+1$ points to its nearest point of Q ; two of them, p_i and p_j , share a nearest point q , and $\rho_1(P_n) \leq d(p_i, p_j) \leq d(p_i, q) + d(q, p_j) \leq 2\rho_1(Q)$. $\square$

Proposition 5 is the obstruction. Proposition 6 bounds the contact count from below, and Proposition 7 says the simple placement rule we use loses at most a factor two in radius against any

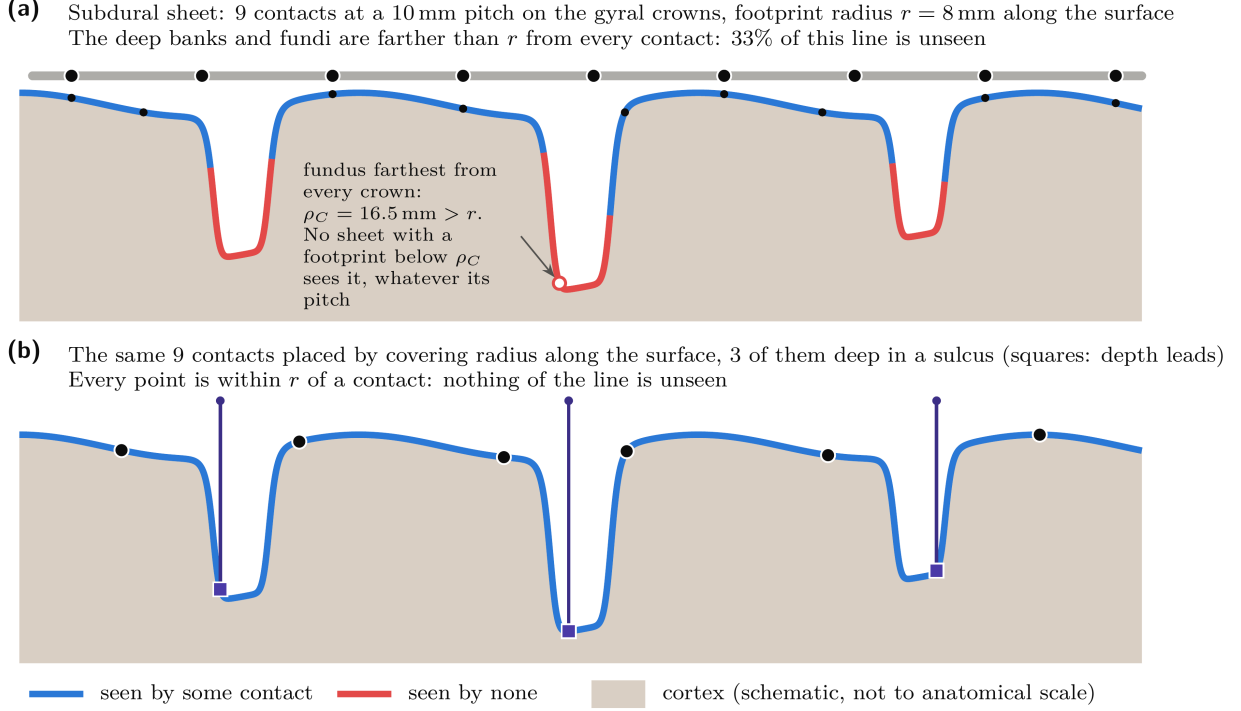


FIGURE 3. The site floor in one cross-section (schematic; distances are arc length along the pial line). (a) A subdural sheet with 9 contacts at a 10 mm pitch; large dots are contacts on the sheet and small dots the crown points they rest on. With footprint radius $r = 8$ mm the deep banks and fundi are seen by no contact, 33% of the line. The fundus marked is $\rho_C = 16.5$ mm from every crown point, so no sheet with a footprint radius below that sees it, at any pitch (Proposition 5). (b) The same 9 contacts placed by covering radius along the line, 3 of them deep in a sulcus on depth leads: every point is within r of a contact. Geometry from `tikz/fold_geometry.py`.

placement with the same number of contacts. On a surface the count scales like the inverse square of the radius, so the factor two in radius becomes about four in contact count.

4. DESIGNING THE ARRAY

The design procedure follows from Section 2 and is summarised in Fig. 4. Given M , X , C , r and k : compute the site floor ρ_C by one multi-source Dijkstra run from every admissible site. If $\rho_C \geq r$ the sites are infeasible (Proposition 5), and C has to be enlarged, with depth leads or strips placed into the sulci, or a larger radius accepted. Otherwise add contacts by farthest-point sampling, updating $\rho_1, \dots, \rho_k$ exactly after each, until $\rho_k < r$. Then check the design on the mesh: $X \subseteq R_k$ triangle by triangle, the certificate of the covered region with its disk test, and the capture curves W_1 , W_k and L . The procedure needs only the patient's pial surface, which is part of standard presurgical imaging [5, 11], and is exact on the mesh at every step.

5. RESULTS ON A TEMPLATE CORTEX

Surface and territory. The surface is the left pial surface of the colin27 template [16], as distributed with FieldTrip [25]: 175 409 vertices and 350 814 triangles, every edge in exactly two triangles, Euler characteristic 2. The territory X is every triangle within 35 mm, along the surface, of

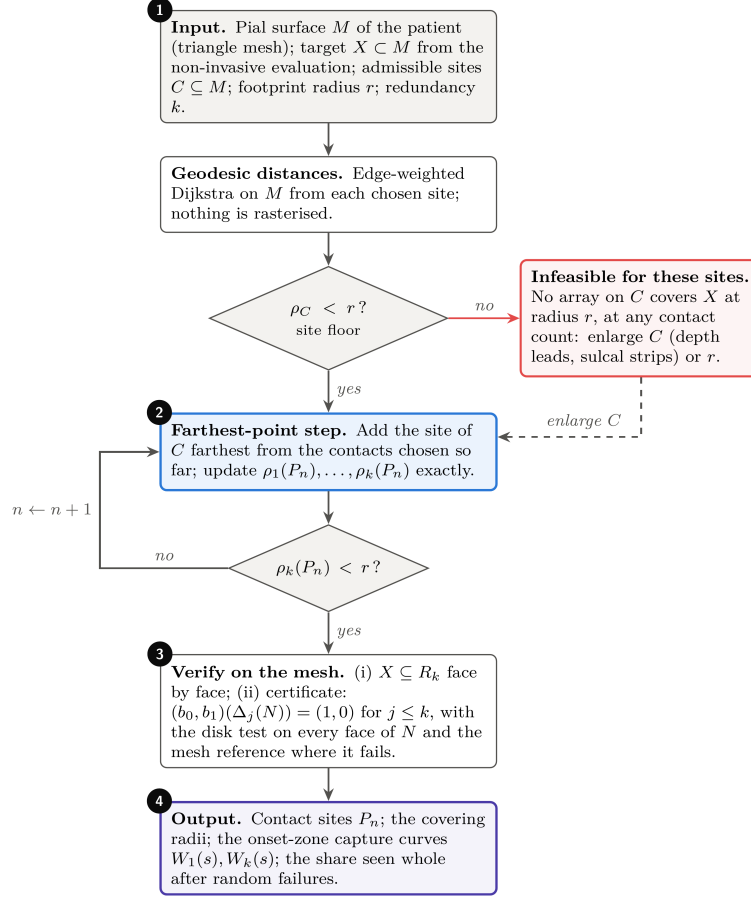


FIGURE 4. The design procedure. The site-floor test (Proposition 5) decides before any placement whether the admissible sites can cover X at radius r . Farthest-point steps (Proposition 7) add contacts until the k -th covering radius drops below r (Proposition 2); the design is then verified on the mesh and its onset-zone capture is reported (Proposition 4).

a parietal anchor at MNI $(-58, -38, 38)$: 4977 vertices, 9659 triangles and 3777 mm^2 , a topological disk. The reference array is the 8×8 parietal grid of the FieldTrip ECoG tutorial layout [32], 10 mm pitch, transferred to the template by laying the lattice on the envelope of the hemisphere and assigning each lattice point to the nearest gyral vertex; its median neighbour spacing is 9.75 mm in a straight line and 10.24 mm along the surface. This is template geometry, not a patient's implantation.

Where the contacts sit decides the coverage. At $r = 8 \text{ mm}$ the 8×8 grid leaves 2082.0 mm^2 (55.1%) of X unseen, and its covered region, with Betti numbers $(b_0, b_1) = (9, 2)$, is in pieces and has holes. The same 64 contacts placed by farthest-point sampling over X leave 1.0 mm^2 unseen (Figs. 5 and 7b). At 10 and 12 mm the grid still misses 41.3% and 28.6%, and the placed array misses nothing. The covering radius says why. The grid's is 21.57 mm, the placed array's 6.99 mm (Fig. 6). The contact count is the same. Only the positions differ.

The site floor on real anatomy. Take C to be the gyral part of X , the 3016 vertices of negative sulcal depth, where a sheet on the envelope puts its contacts. One multi-source Dijkstra run gives

(a) the published 8×8 grid, $r = 8$ mm

(b) the same 64 contacts, placed by radius

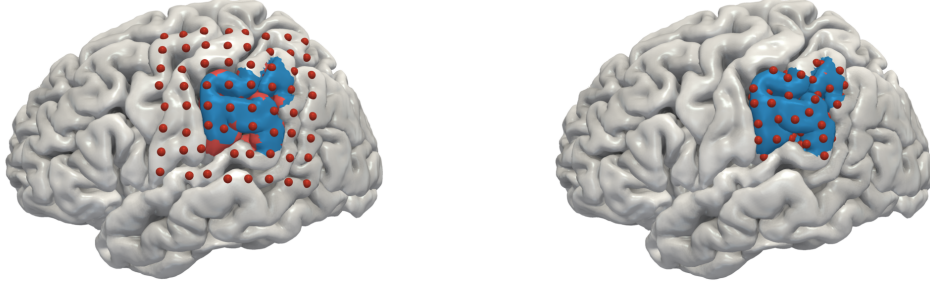


FIGURE 5. The target territory X on the colin27 left hemisphere at footprint radius $r = 8$ mm, blue where some contact sees the cortex and red where none does. (a) The documented 8×8 grid (red spheres), laid on the envelope of the hemisphere: most of the unseen cortex lies on sulcal banks, hidden from this view. (b) The same 64 contacts placed by covering radius. Rendered from the repository's result files with PyVista.

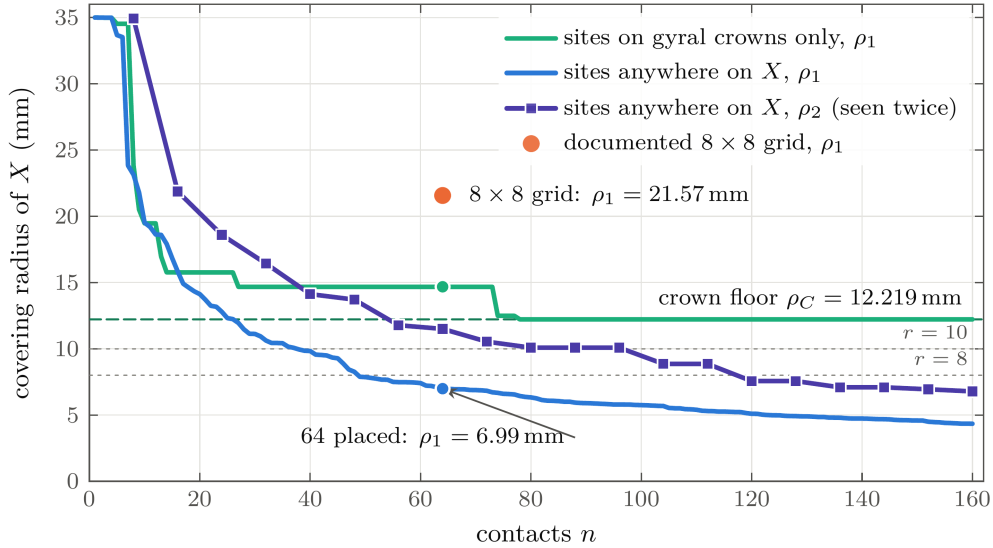


FIGURE 6. Covering radius of X against contact count, by farthest-point sampling. Blue: sites anywhere on X . Aqua: sites on gyral crowns only, which cannot fall below the crown floor $\rho_C = 12.219$ mm (dashed). Violet: the second covering radius ρ_2 , which sets double coverage. Orange: the documented 8×8 grid. Dotted lines mark $r = 8$ and 10 mm; a design covers X at radius r once its curve is below the line (Proposition 2).

$\rho_C = 12.219$ mm, attained at MNI $(-51.1, -42.2, 22.1)$ part way down a sulcal bank. Farthest-point sampling restricted to the crowns reaches that floor at $n = 78$ and stays there (Fig. 6). By Proposition 5, no array with its contacts on the crowns covers X at a footprint radius below 12.219 mm, whatever its pitch or contact count. With 64 crown contacts at $r = 8$ mm, 223.7 mm^2 of X stays unseen.

How few contacts. At each footprint radius, Table 1 gives the smallest n for which the placed contacts leave nothing of X unseen and the certificate gives $(b_0, b_1) = (1, 0)$ with the mesh reference

TABLE 1. Fewest contacts placed by covering radius for which nothing of X is unseen, with the certificate of the covered region R_1 (nerve and mesh agree at every row), the packing lower bound of Proposition 6, and the number of disk-test failures among the nerve’s faces, where the mesh decides.

r (mm)	n	ρ_1 (mm)	unseen	(b_0, b_1)	lower bound	disk-test fail.
14	22	13.25	0	(1,0)	8	8
12	30	11.12	0	(1,0)	8	10
10	46	8.88	0	(1,0)	12	32
8	66	6.95	0	(1,0)	16	24

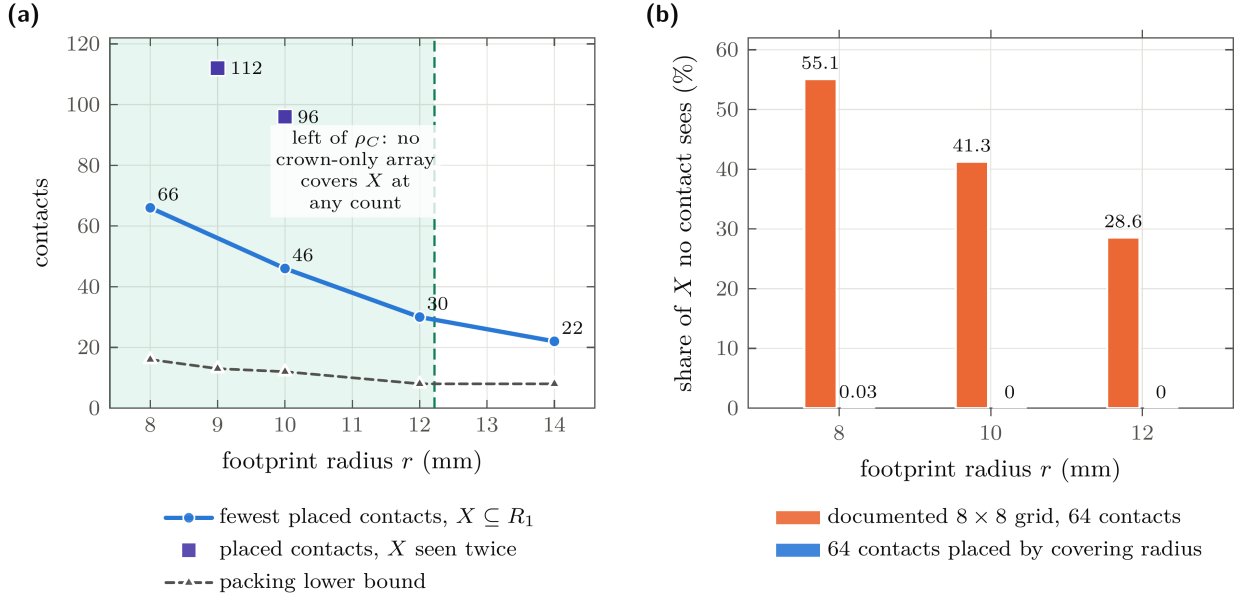


FIGURE 7. (a) Fewest contacts placed by covering radius that see all of X (blue), and that see all of X twice (violet), against the packing lower bound (grey). Left of the crown floor (shaded) no array with contacts on the crowns covers X at any count. (b) The share of X that no contact sees, for the documented 8×8 grid and for the same 64 contacts placed by covering radius.

agreeing. From $r = 14$ to 8 mm the count rises from 22 to 66, against packing lower bounds of 8 to 16 (Fig. 7a). All designs at radii below 12.22 mm need sites off the crowns: 22 of the 66 sites at 8 mm and 14 of the 46 at 10 mm lie in sulcal territory that a sheet cannot reach and a depth lead or a sulcal strip can.

Seeing the onset zone whole. Table 2 and Fig. 8 give the capture of an onset zone of radius s placed uniformly in X . With the 8×8 grid at $r = 8$ mm, an onset zone of radius 5 mm is seen whole in 13.4% of its possible positions and missed entirely in 16.0%. An onset zone of radius 10 mm is never seen whole, and one of radius up to 13.9 mm can lie in X and be missed entirely. Proposition 4(2) predicts at least $\rho_1 - r = 13.57$ mm wherever the grid’s farthest point leaves room for the disk in X . Sixty-four crown contacts never miss a 5 mm onset zone entirely but see it whole in only 77.3% of positions: the rest are seen in part, the case of truncated localisation. The same 64 contacts placed anywhere on X see it whole in 97.3% of positions, the remainder being the positions

TABLE 2. Covering radii and onset-zone capture of each design on X . W_1 : positions in which an onset zone of radius s is seen whole; L : missed entirely; W_2 : seen twice, hence whole after any one failure; h : radius of the largest onset zone that can be missed entirely. Percentages of admissible positions, area-weighted.

design	r (mm)	ρ_1 (mm)	ρ_2 (mm)	unseen (mm ²)	W_1 (%)		L (%) $s = 5$	h (mm)	W_2 (%) $s = 5$
					$s = 5$	$s = 10$			
8×8 grid, 64	8	21.57	25.02	2082.0	13.4	0.0	16.0	13.9	0.0
8×8 grid, 64	10	21.57	25.02	1559.4	28.0	2.1	8.5	11.7	1.1
crown sites, 64	8	14.68	15.67	223.7	77.3	47.4	0.0	4.1	10.1
placed, 64	8	6.99	11.50	1.0	97.3	94.0	0.0	0.0	2.0
placed, 66	8	6.95	10.78	0.0	100.0	100.0	0.0	0.0	2.1
placed, 46	10	8.88	13.71	0.0	100.0	100.0	0.0	0.0	1.1
placed, 96	10	5.79	10.09	0.0	100.0	100.0	0.0	0.0	99.6
placed, 112	9	5.30	8.86	0.0	100.0	100.0	0.0	0.0	100.0

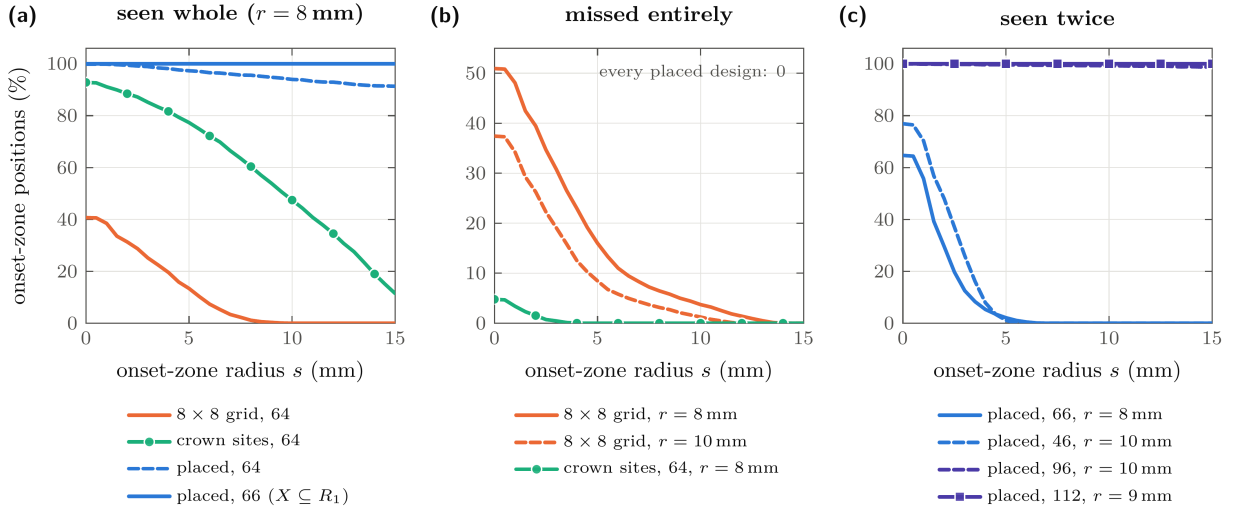


FIGURE 8. Onset-zone capture against onset-zone radius s , over all positions in which the zone fits in X . (a) Seen whole, $W_1(s)$, at $r = 8$ mm: the 8×8 grid, 64 contacts on the crowns, 64 and 66 contacts placed by covering radius. (b) Missed entirely, $L(s)$; every placed design misses nothing. (c) Seen twice, $W_2(s)$, which by Proposition 3 is the share still seen whole after any one contact fails: single-coverage designs (blue) lose it quickly with s , double-coverage designs (violet) keep it. Data from `run_soz_capture.py`.

within 5 mm of the 1.0 mm^2 they leave unseen, and 66 contacts see every onset zone of every size whole, as Proposition 4(1) requires.

When contacts fail. Single coverage is fragile. The 66-contact design sees a 5 mm onset zone twice in only 2.1% of positions, so one well-placed failure can open a gap under most onset zones. Double coverage repairs this. With 96 contacts at 10 mm, all but 1.7 mm^2 of X is seen twice and $W_2(5) = 99.6\%$; with 112 contacts at 9 mm all of X is seen twice by direct computation on the mesh, $\rho_2 = 8.86$ mm, and every onset zone survives any single failure. The nerve certificate of this last design is $(1, 0)$ for $\Delta_2(N)$ but the mesh finds $(1, 1)$: a hole in R_2 outside X , missed by the nerve

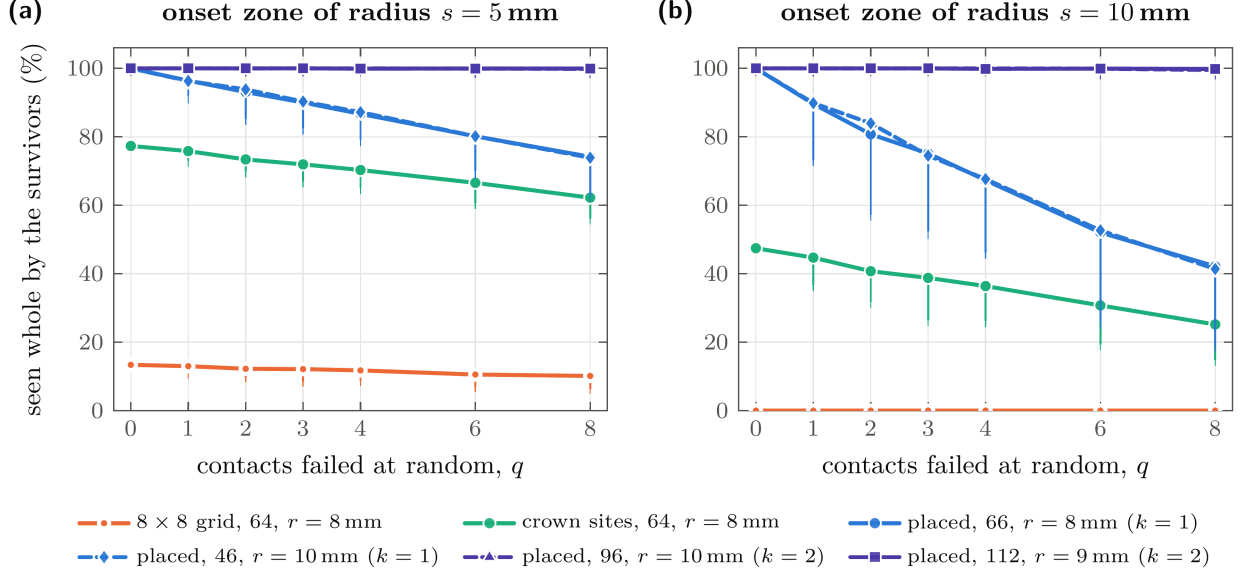


FIGURE 9. Onset zones still seen whole when q contacts fail at random, for onset zones of radius (a) 5 mm and (b) 10 mm. Lines are means over 100 random failure sets for each q ; whiskers reach down to the 5th percentile. Single-coverage designs ($k = 1$, blue) degrade with every failure; double-coverage designs ($k = 2$, violet) stay near 100% through eight failures. The uniform failure model is an assumption; failures clustered in space would be harder on every design.

because 961 of its 16835 faces fail the disk test. It does not affect X , and it is why containment is always decided on the mesh.

Under random failure of q contacts (100 draws for each q), the share of 5 mm onset zones still seen whole falls, for the 66-contact single-coverage design, to 96.4% at $q = 1$, 86.7% at $q = 4$ and 74.2% at $q = 8$ (5th percentile over draws 63.1%). For the double-coverage designs it stays at or above 99.8% throughout (Fig. 9). The 8×8 grid starts at 13.4% and ends at 10.1%. The price of failure-proof sampling is contacts: 112 at 9 mm against 66 at 8 mm, which is the trade the design procedure makes explicit.

6. DISCUSSION

What optimal coverage buys. The results separate two things that “coverage” runs together: how many contacts an array has, and where they sit. On this territory the documented grid and a placed array with the same 64 contacts differ by a factor of three in covering radius, and by more than half the territory in what they see. The covering radius explains the difference in one number, predicts the largest onset zone an array can miss (Proposition 4), and is cheap enough to optimise directly. Pushing ρ_k below r gives a worst-case guarantee that no average-case measure gives: every onset zone of every size in X is seen whole, and stays so after any $k - 1$ failures (Proposition 3).

The site floor explains why subdural sheets have limits that no pitch or contact count removes (Proposition 5). Between 28% and 33% of the contacts of the designs below the floor sit in sulcal territory, so an optimal design for a folded territory is in part a depth-lead design. Stereo-EEG and hybrid implantations reach sulcal and deep cortex that subdural grids do not [19, 26]. The procedure here says, for a given patient and territory, how much of the territory a sheet can see at all, before anything is implanted.

What it does not buy. Seeing the onset zone whole is a condition for localising it, and localising it is a condition for resecting it. Neither is sufficient for seizure freedom. The recorded onset zone may not be the whole epileptogenic zone, the territory X may be wrong, and eloquent cortex may forbid the resection. Whether better coverage changes outcome is therefore an empirical question, and the published evidence does not settle it: the completeness of resection of the recorded onset zone was not associated with outcome in one cohort [12], while spatial perturbation of implantations has been proposed as a way to test whether the epileptogenic zone was sampled at all [18].

A test of clinical relevance. The test needs patients with the reconstructed pial surface, the implanted contact coordinates, the resection and the surgical outcome, as in open intracranial datasets such as OpenNeuro ds003029 [23]. For each patient, compute on the patient’s own surface the covering radius of the resected territory by the implanted contacts, W_1 and L over it, and the same for the design this paper’s procedure would have produced with the same number of contacts. Then test, under a plan fixed before any outcome label is read, whether coverage of the resected territory predicts seizure freedom. A null result would bound the clinical value of optimal coverage. A positive one would justify a prospective comparison of designed and conventional implantations.

Limitations. Everything here is computed on one template hemisphere and one territory, not on patients. The footprint radius r is a modelling parameter, the distance along the cortex over which a contact is credited with recording an onset; it is swept, not measured, and calibrating it from lead fields or from recorded spread is the most important extension. Geodesic distance is shortest-path distance on the mesh edges, which slightly overestimates the true surface distance. Contact sites are mesh vertices, a site off the crowns stands for a depth or sulcal contact without modelling its trajectory, and surgical constraints (vessels, eloquent cortex, safe entry points) are not modelled; they enter the procedure as a smaller site set C , which Proposition 5 then prices. The uniform priors over onset position and over failures are assumptions. Failures clustered in space, as real contact failures on scalp arrays have been found to be [4], would be harder on every design, and would reward coverage deeper than $k = 2$.

7. METHODS

Surface and distances. The left pial surface of colin27 is taken from FieldTrip at commit 8e2307d, in MNI millimetres; it is a de-identified reference anatomy, not patient data. Distances are Dijkstra distances [8] on the edge graph with Euclidean edge lengths. A triangle belongs to the footprint of a contact when all three of its vertices lie within distance r of it; the unseen area of X is the area of its triangles in no footprint. Gyral vertices are those of negative FreeSurfer sulcal depth.

Designs. Farthest-point sampling starts from the vertex of X nearest to the anchor and adds, at each step, a candidate farthest from the contacts so far, with candidates all vertices of X (placed designs) or its gyral vertices (crown designs); ρ_1 is recorded exactly at every step, and ρ_k every eight steps. The smallest designs of Table 1 are found by scanning n upwards. Packing bounds come from a maximal $2r$ -separated subset of X . The certificate is computed with the nerve of the footprints, the Betti numbers of $\Delta_k(N)$ over $\text{GF}(2)$, a disk test on every face of the nerve, and a mesh reference for R_k computed without the nerve, all as in [4].

Capture and failure. For each design, one multi-source Dijkstra run from the vertices of the triangles of X held by fewer than k footprints gives, at every vertex, the distance to the set $M \setminus R_k$, and one from the vertices of the seen triangles gives the distance to R_1 . By Proposition 4, W_1 , W_2 and L follow by thresholding at each s from 0 to 15 mm in steps of 0.5 mm, with vertex areas (a third of the incident triangle areas) as the measure and centres admissible when their distance to the outside of X exceeds s . For contact failure, $q \in \{1, 2, 3, 4, 6, 8\}$ contacts are removed uniformly at random, 100 times for each q with fixed seeds, the survivors’ footprints are recomputed, and W_1

is taken again at $s = 5$ and 10 mm. The analysis is `run_soz_capture.py`; it runs in about twenty seconds.

Figures. Figures 1–4 and 6–9 are TikZ and pgfplots sources, drawn from the result files by `make_ecog_paper_data.py` and rendered at 600 dpi; the colours are a single palette checked for colour-blind separation, with markers and line styles as a second channel. Figure 5 is the PyVista render of the repository.

DATA AVAILABILITY

No patient data are used. The template surface, the designs (every contact by mesh vertex and MNI coordinate) and all result files are in the repository below.

CODE AVAILABILITY

All code, result files and figure sources are in the repository <https://github.com/creelie/kshadow-coverage> [4]. This manuscript is `paper/ecog/ecog_coverage.tex`, built by `paper/ecog/build.sh`. Every number in it is a macro generated from the result files.

AUTHOR CONTRIBUTIONS

The idea of certifying k -fold sensor coverage through iterated nerve complexes was first developed by D.B. with U.B. [2]. A.B. extended it to electrode arrays on the cortex. D.B., U.B. and A.B. wrote the code and carried out the analyses together, and all three authors wrote the paper.

USE OF AI TOOLS

Drafting of this manuscript, of its TikZ figures and of the analysis script `run_soz_capture.py` was assisted by an AI system (Claude, Anthropic). The authors are responsible for the content.

COMPETING INTERESTS

The authors declare no competing interests.

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